

ENV 6106

Air Pollution Modeling

Spring 2023

Topics

- AERMAP
- Expectation
 - Understand how AERMAP works and be able to run AERMAP
- Readings: AERMAP User guides

Components of AERMOD

- AERMET: meteorological data preprocessor
 - AERSURFACE: for surface characteristics
 - AERMINUTE: for 1-min ASOS data
- AERMAP: terrain data preprocessor
- AERMOD: uses input data (meteorological, terrain data etc) to perform refined analysis
 - BPIPPRIM: a multi-building dimensions program incorporating the GEP technical procedures for building downwash applications.

AERMAP

- Get terrain elevation data for all receptors and emission sources
 - Receptor: where concentrations will be calculated
 - Different from “receptor” models
- Calculate hill height scale
 - A parameter to determine how plume shifts due to an elevated terrain feature

Terrain data

- Many terrain elevation data available at various resolution
 - 1 meter to dozens of km
- Some terrain data types
 - USGS Digital Elevation Models (DEM)
 - NASA Shuttle Radar Topography Mission (SRTM)
 - Global 30 Arc-Second Elevation (GTOPO30)
 - **National Elevation Dataset (NED)**
 - Generally this is what we should use for AERMOD system
 - 3DEP (3D Elevation Product)

National Elevation Dataset

- NED is high resolution terrain elevation data seamlessly cover entire US, with a consistent datum, projection, and elevation units (meters)
- Data available in ~30, 10, and 3-meters
 - 1, 1/3, and 1/9-arcsecond
- Easier to handle, and works well with AERMAP
- AERMAP only accepts uncompressed GeoTIFF format

Download & convert NED data

- Source of NED data has been changed
- Follow the document below to download NED data
 - https://gaftp.epa.gov/Air/aqmg/SCRAM/models/related/aermap/Access_and_Conversion_of_Elevation_Data_for_AERMAP.pdf
 - 1 & 2 arcsecond data available from EPA FTP server
- If need high resolution, steps:
 - <https://apps.nationalmap.gov/downloader/#/>
 - Zoom to area of interests, draw a rectangle
 - Under “Dataset”, choose “Elevation Products (3DEP)”, choose “1/3 arc-second DEM”, Or “1 arc-second DEM”
 - Click “Find Product”
 - For UCF region, find the file “USGS NED 1/3 arc-second n29w082 1 x 1 degree”, this file contains NED data for areas between 29N-30N, and 81-82W, a 1 degree by 1 degree region. Click “Download”

Download & convert NED data

- Steps:
 - Unzip the file to a folder (e.g: C:\AERMAP)
 - Download GDAL software, and unzip to a folder (e.g: C:\AERMAP\GDAL)
 - Press Windows button and R at the same time, a “Run” windows should pop up
 - Type cmd, and press “OK”
 - Type “PATH
%PATH%;C:\AERMAP\GDAL\bin;C:\AERMAP\GDAL\bin\gdal\apps”
and press enter. Actual folder path could be different
 - Type “gdal_translate -of GTIFF -co COMPRESS=NONE
E:\NED\grdn29w082_13 Path_of_your_choice\output_name”
 - No need to use other software such as ArcMap to do the conversation

How to run AERMAP

- No interactive interface
- Reads “aermap.inp” file, the control file
- The control file also contains some pathways and keywords
- Pathways are only two letters
 - CO: overall job **C**ontrol
 - SO: **S**ource information (optional)
 - RE: **R**Eceptor information
 - OU: **O**utput information

CO pathway

AERMAP CO pathway

- Overall job control
- Start and end CO pathway
 - CO STARTING
 - CO FINISHED
- Between start and finish
 - CO TITLEONE *a_bunch_of_title_info*
 - Information to identify this AERMAP run
 - CO TITLETWO *more_title_info*
 - More information if needed, optional

AERMAP CO pathway

- CO RUNORNOT
 - Do a setup check before model execution?
 - Do check: CO RUNORNOT NOT
 - No check: CO RUNORNOT RUN
- CO DATATYPE
 - Type of raw input terrain elevation data
 - Old DEM data: CO DATATYPE DEM
 - Use NED data: CO DATATYPE NED

AERMAP CO pathway

- CO DATAFILE
 - Name of input raw terrain elevation data
 - Can be repeated to include several files
 - Use NED: CO DATAFILE *ned_file_name*
 - NED default in m, can change the unit if needed
- CO ANCHORXY
 - Client could define own facility center as (0,0)
 - Transform user-defined coordinate system to UTM
 - A simple shift, any chosen point would work

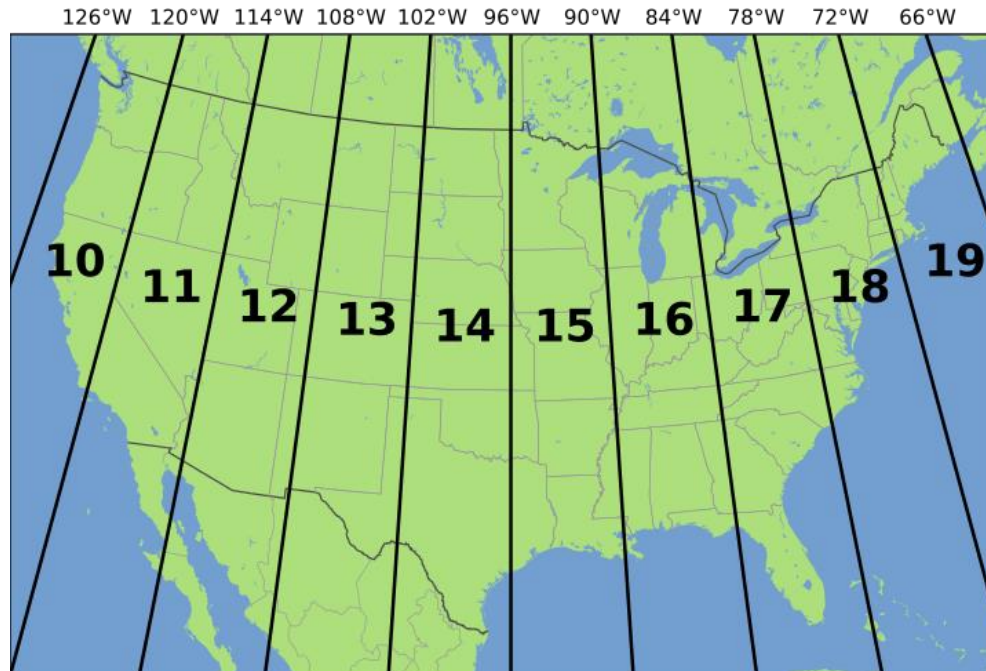
AERMAP CO pathway

- CO ANCHORXY

- ANCHORXY *Xauser Yauser Xautm Yautm Zautm (NADA_value)*
- *Xauser Yauser*: X and Y coordinates of the chosen reference point in user defined system
- *Xautm Yautm* : X and Y coordinates of the same reference point in UTM system
- *Zautm*: UTM zone #
- *NADA_value*: define what datum should be used, chose 4 for NAD83 datum, default for NED data downloaded from MRLC
- See Table 3-3, page 3-17 of AERMAP user guide for more info
- If UTM coordinates is used as user-defined system, then *Xauser=Xautm*, and *Yauser=Yautm*

AERMAP CO pathway

- How to convert lat/long to UTM
 - Many tools available: e.g.:
 - <http://rcn.montana.edu/Resources/Converter.aspx>
 - Note: UTM easting: X; UTM Northing: Y



Example CO pathway

CO STARTING

TITLEONE AERMAP Test case centered at UCF

DATATYPE NED

DATAFILE Test_UCF.tif

ANCHORXY 480415 3163945 480415 3163945 17 4

RUNORNOT RUN

CO FINISHED

SO pathway

AERMAP SO pathway

- Start and end
 - SO STARTING
 - SO FINISHED
- Between start and end
 - SO LOCATION
 - Defines coordinates of emission sources
 - SO LOCATION *Srcid Srctyp Xs Ys (Zs)*
 - *Srcid*: source ID, you name it
 - *Srctyp*: source types: POINT, VOLUME, AREA, AREAPOLY, AREACIRC, OPENPIT, POINTCAP, or POINTHOR
 - *Xs, Ys*: coordinates of source in user-defined system
 - *Zs*: optional, elevation above sea level in meters

AERMAP SO pathway

- SO INCLUDED
 - Include an external file that contains source information (i.e.: many SO LOCATIONS)
 - SO INCLUDED *file_name*
- The entire SO pathway is optional, sometimes elevation at emission source is known, no need to extract from NED data
 - But it's worth to do a double check

Example SO pathway

SO STARTING

LOCATION FAKESRC POINT 480415 3163945

SO FINISHED

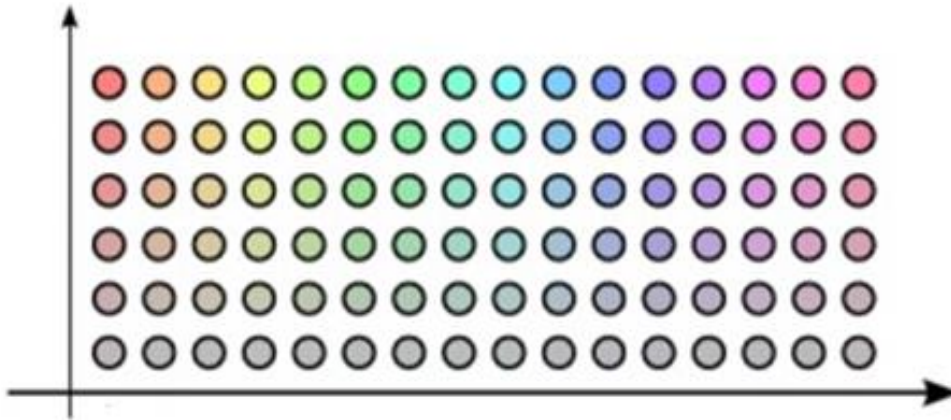
RE pathway

AERMAP RE pathway

- Start and end
 - RE STARTING
 - RE FINISHED
- Between start and end
 - Four ways to specify receptor locations
 - An evenly distributed **Cartesian** network
 - Arbitrarily chosen discrete **Cartesian** points
 - An evenly distributed **polar** network
 - Arbitrarily chosen discrete **polar** points
 - At least one should be included, internal or external

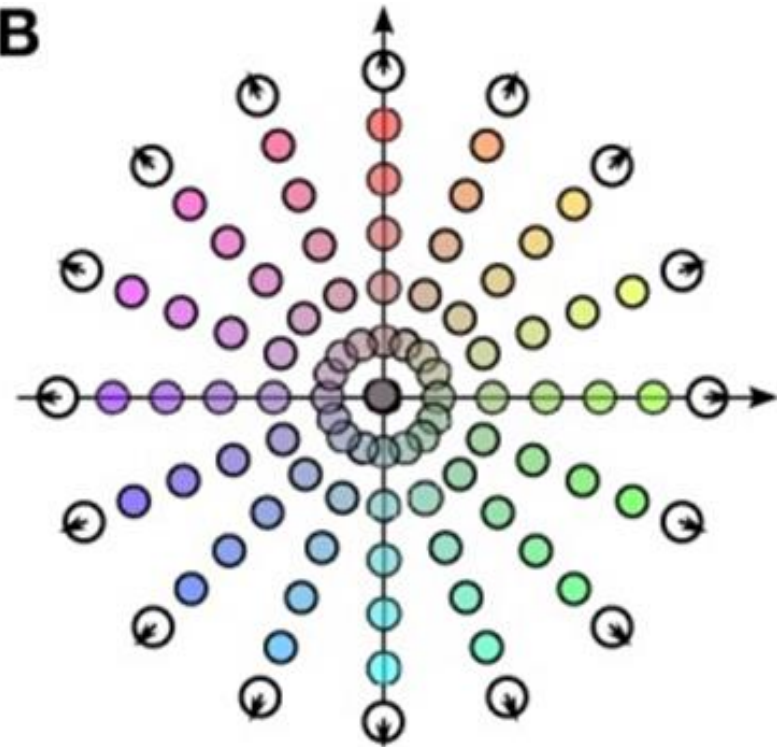
AERMAP RE pathway

A



Cartesian

B



Polar

AERMAP RE pathway

- 1) An evenly distributed **Cartesian** network
 - Method a:

```
RE GRIDCART NetID STA
RE GRIDCART NetID XYINC Xinit Xnum Xdelta Yinit Ynum Ydelta
RE GRIDCART NetID END
```

Or

```
RE GRIDCART NetID STA
                        XYINC Xinit Xnum Xdelta Yinit Ynum Ydelta
                        END
```

NetID: name of the receptor network;

Xinit, *Yinit*: initial coordinate (lower-left corner of receptor network)

Xnum, *Ynum*: number of receptor along X and Y direction

Xdelta, *Ydelta*: spacing between two receptors along X and Y direction

AERMAP RE pathway

- 1) An evenly distributed **Cartesian** network
 - Method b:

```
RE GRIDCART NetID STA
RE GRIDCART NetID XPNTS Gridx1 Gridx2 ..... Gridxm
RE GRIDCART NetID YPNTS Gridy1 Gridy2 ..... Gridyn
RE GRIDCART NetID END
```

Or

```
RE GRIDCART NetID STA
                XPNTS Gridx1 Gridx2 ..... Gridxm
                YPNTS Gridy1 Gridy2 ..... Gridyn
                END
```

NetID: name of the receptor network;

Gridx1... *Gridxm*: X coordinates where you want to place receptor

Gridy1... *Gridyn*: Y coordinates where you want to place receptor

Final receptor network is a combination of all Xs and Ys listed

AERMAP RE pathway

- 2) Arbitrarily chosen discrete Cartesian points
 - RE DISCCART *Xcoord Ycoord (Zelev) (Zflag)*
 - *Xcoord, Ycoord*: X and Y coordinates of receptor
 - *Zelev, Zflag*: elevation for elevated&flagpole receptor
 - Can repeat as much as needed
- We will not go over 3) A evenly distributed polar network, and 4) Arbitrarily chosen discrete polar points here, see section 3.4.2.2 and 3.4.5.1 of AERMAP user guide for more information

AERMAP RE pathway

- Too many receptors? Use external file
 - RE INCLUDED *file_name*
 - External file contains repeated RE XXXX
- Can pick and mix, make several different networks at the same time

Example RE pathway

RE STARTING

GRIDCART REC1 STA

XPNTS 480015 480115 480215

YPNTS 3163545 3163645

GRIDCART REC1 END

RE FINISHED

OU pathway

AERMAP OU pathway

- Start and end
 - OU STARTING
 - OU FINISHED
- Between start and end
 - OU RECEPTOR *out_receptor_file_name*
 - OU SOURCLOC *out_source_file_name*
 - OU SOURCLOC is Optional

Example OU pathway

OU STARTING

RECEPTOR TEST_UCF_RECEPTOR.TXT

SOURCLOC TEST_UCF_SOURCE.TXT

OU FINISHED

AERMAP output

- AERMAP.OUT: a log file
- MAPDETAIL.OUT
 - Information on input raw terrain data.
Resolution, extend etc
- MAPPARAMS.OUT
 - Properties of input raw terrain data file
- Others output also possible depends on options chosen (e.g., choose a subset of the input raw data etc)

Chose AERMAP domain

- No one-size-fits-all solution for best size
- Consider extend ~10km outside all receptors
 - Terrain outside receptor network still impact hill height scale of nearby receptor
- Do include terrain features that would impact plume positions
 - E.g. A nearby big hill